

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template AT

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

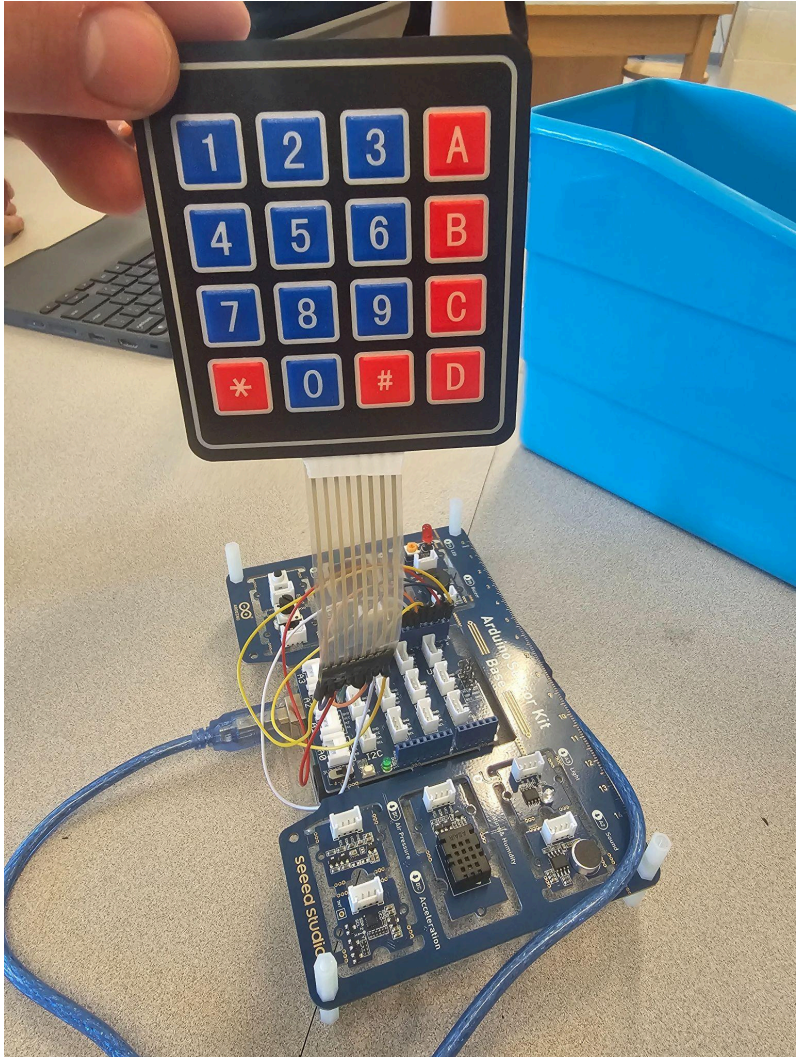
Design Idea #1	Arduino with card reader
Date:	4/19
Description:	Using the Arduino as a hub, create a system that revolves around a card reader.
Drawing that includes distinguishing information such as scale, dimensions, and materials:	N/a

Design Idea #2	Arduino with keypad
Date:	2/23

Name: _____

Date: _____

Class: _____

Description:	<i>This idea is far more cost-effective than using a numerical algorithm that uses the keypad to interact with the system</i>
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

Name: _____

Date: _____

Class: _____

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<i>Design Idea #3</i>	
Date:	
Description:	
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

<i>Design Idea #4</i>	
Date:	
Description:	
Drawing that includes distinguishing information	



Name: _____

Date: _____

Class: _____

such as scale, dimensions, and materials:	
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Design Idea #5 (if needed)	
Date:	
Description:	
Drawing that includes distinguishing information such as scale, dimensions, and materials:	

Design Idea Comparison and Selection (D1, D2)

(List your requirements for Element C below.)	Weight (multiplied by each design score)	Design 1	Design 2	Design 3	Design 4	Design 5

Name: _____

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Requirement A	Cost-effective	no	yes			
Requirement B	Efficient	yes	yes			
Requirement C	Easy to use	yes	somewhat			
Requirement D	Tracks time	yes	yes			
Requirement E	Tracks amount	yes	yes			
Total score						

Our team chooses design _____ 2 _____ to solve the problem because
_____ It fulfills all the
requirements _____

Design Blueprint (D4)

Design Blueprint #1

Date:

Sketch or CAD Drawing (orthographic/multiview):

Manufacturing Methods (cut, saw, paste, screw, tie, glue):

Step 1:

Step 2:

Step 3:

Supply List and Costs:

Name: _____

Date: _____

Class: _____

Item	Cost each	# needed	Total
Arduino	N/a	1	N/a
Wires	N/a	1	N/a
Keypad	N/a	1	N/a
Total Cost	N/a	N/a	N/a

Plan of Action (D3)

Iterations for the Solution (How did you change your prototype after building and testing it?):

Date of Iteration: 2/24

Description of Iteration: I created the original code and ran the program

Images if applicable: N/a

Date of Iteration: 4/25

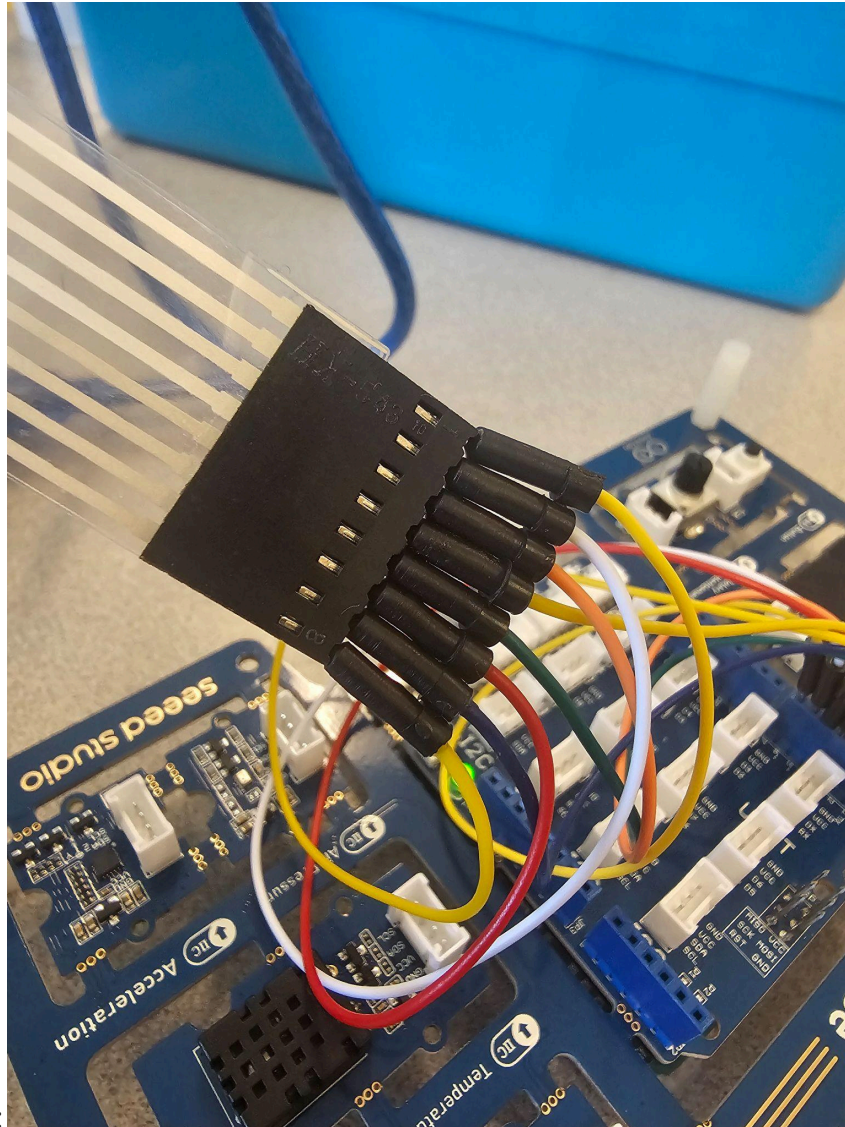
Description of Iteration: Fully attached the keypad and reorganized the wires.



Name: _____

Date: _____

Class: _____



Images if applicable:

Date of Iteration: N/a

Description of Iteration: N/a

MyDesign Portfolio - Element D Initial Template

Name: _____

Date: _____

Class: _____

Images if applicable:N/a

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for Generating Possible Solutions _____.	was <i>comprehensive,</i> <i>iterative,</i> <i>and</i> <i>consistently</i> defensible (viable design highly likely)	was <i>thorough,</i> <i>iterative,</i> <i>and</i> generally defensible (viable design likely)	was <i>adequate,</i> <i>generally</i> iterative and defensible (<i>viable design</i> <i>is possible</i>)	was <i>partial or</i> <i>overly general</i> <i>and only</i> <i>somewhat</i> iterative and/or defensible (raising issues with the design viability)	was <i>incomplete</i> <i>or minimally</i> <i>iterative</i> <i>and/or</i> <i>defensible</i> (<i>raising</i> <i>issues with</i> <i>the design</i> <i>viability</i>)	<i>was not evident</i> <i>that there was</i> <i>an attempt</i>
2. Design Solution Choice. The Design Solution chosen _____.	was <i>well-justified</i> and demonstrated attention to <i>all</i> design requirements	was <i>justified</i> and demonstrated attention to <i>most if</i> <i>not all,</i> design requirements	was <i>explained</i> <i>with</i> <i>reference to</i> <i>at least some</i> design requirements	was <i>not</i> <i>sufficiently</i> <i>explained with</i> <i>reference to</i> design requirements	<i>lacked</i> <i>support</i> <i>related to</i> design requirements	<i>was not</i> <i>attempted or</i> <i>evidence not</i> <i>provided</i>
3. Plan of Action. The plan of action _____.	<i>has</i> <i>considerable</i> <i>merit and</i> <i>would easily</i>	<i>would</i> support repetition and testing	<i>might not</i> <i>clearly or</i> <i>fully support</i> <i>repetition</i>	<i>has insufficient</i> <i>detail to allow</i> <i>for testing for</i> <i>replication of</i>	<i>is lacking to</i> <i>the point</i> <i>that the</i> <i>design</i>	<i>cannot be</i> <i>executed or</i> <i>does</i> <i>not exist</i>



Name: _____

Date: _____

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	support repetition and testing for effectiveness s by others	for effectiveness s by others	<i>and testing for effectiveness by others</i>	<i>results</i>	<i>solution cannot be tested</i>	
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